VAYU | YOUR PERSONAL CROP GUIDE

Lemongrass (citral oil)

How to grow it on your land

Prepared for: General guide (no location chosen)
Prepared on 2026-11-11

Hardy, grows on poor and degraded land and animals leave it alone. The catch is the market: income comes from selling herb to a distillery or distilling oil yourself.

DIFFICULTY	MONEY NEEDED	RISK	FIRST RETURN
2 of 5	**Medium**	**Medium**	**first cut about 90 days after planting**

How to read this guide: [S1] marks a line backed by a source listed on the last page. (general knowledge) marks a line that has not been verified against a source. Lines starting **For your land** use the numbers for your location.

In this guide

- Your priorities: what to do first, in order
- Your planting plan: how much to buy and prepare for your plot
- How to grow it: planting, water, nutrition, pests and diseases, monitoring, harvest, risks
- If your conditions are different, and the sources behind every line

Your land at a glance

Your priorities

Ordered by what matters most on your land.

1. Get your soil tested
Satellite and model data estimate soil pH and texture only roughly, and cannot see nutrients, salinity or soil health. Get a proper soil test (pH, organic carbon, nutrients, salinity) from your KVK or the state soil-testing laboratory; the Soil Health Card scheme provides this. (general knowledge)

2. Talk to your KVK first
Take this guide to your nearest Krishi Vigyan Kendra and ask for their local advice and trusted planting material. (general knowledge)

3. Start small
Plant or build a small trial first. Learn the crop on your own land, then expand from what actually worked. (general knowledge)

4. Line up a buyer before you scale
Know who will buy, in what form and grade, and how it will reach them. (general knowledge)

Your planting plan for 1 acre

What	How much	Basis
Hills at 0.45 m x 0.45 m	**about 19,984**	your plot size x cited amounts [S4]
Root slips (2-3 per hill)	**39,968 to 59,952 slips**	your plot size x cited amounts [S1]
Farmyard manure or compost before planting (about 10 tonnes per hectare)	**4.0 tonnes**	your plot size x cited amounts [S1]

Sums of the cited per-plant and per-hectare amounts for your plot size. Real needs vary with soil and variety; confirm with your KVK.

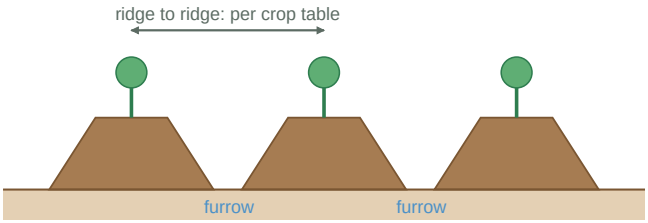
Planting material

- Plant root slips (2-3 tillers each, 25-30 cm long, old roots clipped) from healthy, well-grown clumps. Slips give better oil quality than seed. [S1, S5]
- Recommended varieties include Krishna, Pragati, Nima, CKP-25 and OD-19; Krishna suits irrigated and rainfed land. CSIR-CIMAP (Lucknow) and its Aroma Mission supply slips and advice. [S4, S5, S6]

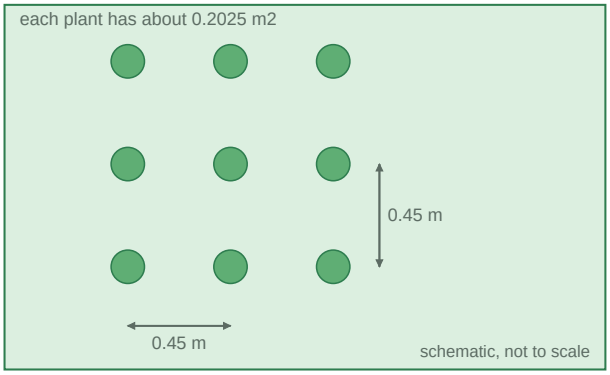
Land preparation and planting

- Plant June-July (last week of May to first week of June in some regions). Open ridges 60 cm apart and plant 2-3 slips per hill. [S1, S4]
- Spacing 45 x 45 cm in plains, or 60 x 45 cm on slopes; 60 x 60 cm was used in an ICAR-IISWC trial. [S4, S7]

- Mix farmyard manure or compost (about 10 tonnes per hectare) into the soil before planting. [S1]



Planting on ridges



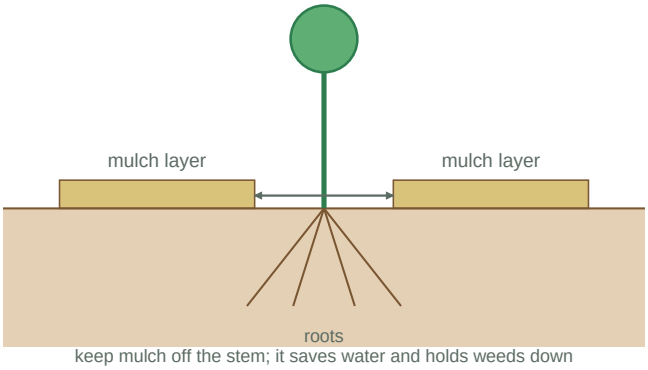
Plant spacing, top view

Irrigation

- Supplemental irrigation in drier areas and in dry spells. [S1]
- Irrigated crops give more oil than rainfed crops. (general knowledge)

Nutrition (organic)

- Farmyard manure or compost at planting and each year. Spent distilled grass applied as mulch (about 3 tonnes per hectare) controls weeds and keeps soil moist. [S1, S4]

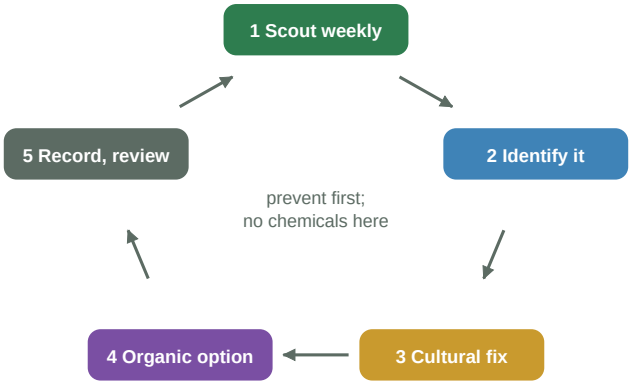


Mulch around a plant

Pests and diseases (organic and cultural only)

- Keep the crop weed-free in the early stage; mulch with spent grass. [S4]
- Lemongrass has relatively few serious pests; scout for leaf spots in humid weather and remove badly affected clumps. (general knowledge)

- For anything beyond these organic and cultural steps, ask the nearest Krishi Vigyan Kendra (KVK) or state agricultural university for their current recommendation. (general knowledge)



Pest and disease loop: prevent first, then look, fix, review

Monitoring

- Check clump health and weed pressure monthly; record each cutting date and the herb weight. (general knowledge)
- Oil recovery rises and falls with season and cutting age, so keep records to find your best cutting time. [S4]

Harvest, yield and post-harvest

- First harvest about 90 days after planting, then every 75-90 days. Cut 10-15 cm above ground, ideally early morning. [S2]
- Herb yield reported: about 215 quintals per hectare in year 1, 360 in years 2-3, 215 in year 4; oil roughly 150 / 250 / 250 / 150 kg per hectare at 0.7 percent recovery. Recoveries between 0.2 and 1.0 percent are reported. [S4, S2]
- Wilt the herb 6-8 hours and distil soon after harvest; steam distillation takes about 3-4 hours in a field unit. [S6]

Risks and costs

- A distillation unit is the main capital cost; many growers sell herb to an existing distiller or join an aroma cluster. (general knowledge)
- Oil prices change with demand; confirm a buyer and current rates before you plant. (general knowledge)

If your conditions are different

What to do if your land, or the season, turns out different from the numbers above.

- **Heavy soil** (fixable): Waterlogged soil should be avoided. On heavy clay plant on ridges, keep drains open, and add compost. [S1, S4]
- **Low rainfall** (fixable): Lower rainfall can be grown with supplemental irrigation (TNAU). Expect a lower oil yield without it, and plan drip or basin irrigation. [S2]
- **Too much rain** (fixable): Waterlogged soil should be avoided. Plant on ridges and keep drainage open. [S1, S4]
- **Soil pH** (fixable): Aim for about pH 5.0-7.5 (6.0 is called suitable by TNAU). Avoid calcareous soil; add compost. [S1, S2]

- **Cold weather** (fixable): It is a warm-climate grass; protect young clumps from frost or choose a warmer site.
  (general knowledge)

About chemicals

This guide gives organic and cultural methods only. It does not name or dose pesticides, fungicides or chemical fertilisers; for those, follow your nearest Krishi Vigyan Kendra (KVK) or state agricultural university and the label of any registered product.

Sources

[S1] https://static.vikaspedia.in/media/files_en/agriculture/crop-production/lemon-grass.pdf
[S2] https://agritech.tnau.ac.in/horticulture/horti_aromatic crops_lemongrass.html
[S4] https://eagriorg.github.io/eagri50/HORT282/pdf/lec08.pdf
[S5] https://eagriorg.github.io/eagri50/HORT282/lec08.html
[S6] https://aromamission.cimap.res.in/aromamission/crop-covered-details/MjQ=?theme=
[S7] https://www.frontiersin.org/articles/10.3389/fagro.2025.1657329/full

Advisory aid from published sources and general agronomic knowledge, not a guarantee of yield or income and not a substitute for local extension advice. Lines marked 'general knowledge' are not verified against a source.